



Linear Control Systems 25411

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Assignment 3

Fall 1404

Due Date: 1404/9/21

Note 1: *purple* problems are bonus ones.

1 Routh-Hurwitz Criterion

The transfer function relating the output engine fan speed (rpm) to the input main burner fuel flow rate (lb/h) in a short takeoff and landing (STOL) fighter aircraft, ignoring the coupling between engine fan speed and the pitch control command, is

$$G(s) = \frac{1.3s^7 + 90.5s^6 + 1970s^5 + 15000s^4 + 3120s^3 - 41300s^2 - 5000s - 1840}{s^8 + 103s^7 + 1180s^6 + 4040s^5 + 2150s^4 - 8960s^3 - 10600s^2 - 1550s - 415}$$

1. Find out how many poles are in the right half-plane, in the left half-plane and on the $j\omega$ -axis.
2. Find out how many poles are in the right half-plane of the axis $\text{Re}\{x\} = 1$, in the left half-plane of the axis $\text{Re}\{x\} = -1$ and between these two axes.
3. Is this open-loop system stable?
4. Is the closed-loop system (with negative unity feedback) stable?

2 Range of a Gain for Closed-Loop Stability

The closed-loop transfer function of a system is

$$G(s) = \frac{s^2 + K_1s + K_2}{s^4 + K_1s^3 + K_2s^2 + 5s + 1}$$

Determine the range of K_1 in order for the system to be stable. what is the relationship between K_1 and K_2 for stability?

3 Evaluating Stability Condition Using Impulse Response

Consider the set of bounded continuous-time signals. The l_∞ norm for this set is defined as follows:

$$\|x\|_\infty = \sup_{t \geq 0} |x(t)|$$

System S is an arbitrary LTI system with impulse response $h(t)$, and $\|S\|$ is defined as follows:

$$\|S\| = \sup_{\|u\|_\infty \neq 0} \frac{\|y\|_\infty}{\|u\|_\infty}$$

a) Prove the following equation:

$$\|y\|_{\infty} \leq \|u\|_{\infty} \int_0^{\infty} |h(\tau)| d\tau$$

b) Prove the following equation:

$$\|S\| = \int_0^{\infty} |h(\tau)| d\tau$$

Hint: find a signal that satisfies the equality (=) in part a.

c) By calculating the corresponding integral for each of the following systems, determine whether the system is stable or not (For a stable system S_i we have $\|S_i\| < \infty$).

1) $S_1 : G_1(s) = \frac{1}{s+1}$

2) $S_2 : G_2(s) = \frac{1}{s-1}$

3) $S_3 : G_3(s) = \frac{1}{s^2+1}$

4) $S_4 : G_4(s) = \frac{1}{(s+1)^2}$

5) $S_5 : h_5(t) = \frac{1}{t+1}$

6) $S_6 : h_6(t) = \frac{1}{t^2+1}$

For each of the systems in part c, generate the step response plots to confirm your solution.

4 Step Response Characteristics

Second-order underdamped systems transfer function can be represented as below:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad \omega_n > 0, \quad 0 < \zeta < 1$$

So, the Laplace transform of the output to the unit step input $u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$ is:

$$Y(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

and

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$$

where $\omega_d = \omega_n \sqrt{1-\zeta^2}$ and $\cos(\phi) = \zeta$.

a) Prove the following equation:

$$T_s \approx \frac{4}{\zeta\omega_n}$$

where settling time T_s is time to go from zero to within 2% of steady state. For which interval of ζ the equation is valid? How can we rectify the equation for $\zeta = 1$?

b) Prove the following equation:

$$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

where overshoot $M_p = \max(y(t)) - 1$. For which interval of ζ the equation is valid? For a fixed ω_n , how can we get the fastest response without overshoot?

MATLAB Assignments

5 System Identification Toolbox

- a) Using *Data.mat* file, derive the step response data ($u(t)$ and $y(t)$) of the unknown plant.
- b) Apply the System Identification toolbox (type *ident* in command line) to identify first order transfer function $G(s) = \frac{k}{s+a}$ using the data set obtained from part a.
- c) Use the sinusoidal input signal and response from *Data2.mat* and repeat part b.
- d) Use white noise input signal and response from *Data3.mat* and repeat part b.
- e) Use lowpass filter $F(s) = \frac{1}{as+1}$ to reduce the effect of noise and apply the System Identification toolbox to identify first order transfer function $G(s) = \frac{k}{s+a}$.
- f) Compare all of the results and suggest the best method and model.
- g) Use the best method to identify second-order transfer function $G_2(s) = \frac{k}{s^2 + a_1s + a_0}$.